



IT-SIMPLERA INSTITUTE

Network Administration Internship Program

WEEK 02 TASK REPORT

IP Addressing (FLSM & VLSM) and VLAN Configuration

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Introduction:

This report documents the Week 2 internship task covering Fixed Length Subnet Masking (FLSM), Variable Length Subnet Masking (VLSM), and Virtual LAN (VLAN) configuration. It includes worked subnetting problems, VLSM design scenarios, and Cisco IOS configuration steps used to implement and verify VLANs in Cisco Packet Tracer and GNS3.

Section 2: FLSM aur VLSM Concepts

2.1 Fixed Length Subnet Masking (FLSM)

FLSM divides a network into subnets that are all the same size, using a single subnet mask across the entire address block. It is simple to plan and document, but it wastes addresses when different departments or links need very different numbers of hosts, because every subnet is sized for the largest requirement.

2.2 Variable Length Subnet Masking (VLSM)

VLSM allows each subnet to use a different mask, sized according to its actual host requirement. Subnets are allocated largest-first from the available address block, which minimizes wasted addresses and is the standard approach used in real enterprise and ISP networks.

2.3 FLSM vs VLSM Comparison

Aspect	FLSM (Fixed Length)	VLSM (Variable Length)
Subnet sizing	All subnets are the same fixed size	Each subnet is sized according to its actual host requirement
Calculation basis	Based on number of subnets required ($2^s \geq \text{subnets}$)	Based on number of hosts required per subnet ($2^n - 2 \geq \text{hosts}$)
Mask formula	New mask = original prefix + borrowed bits (s)	New mask = 32 – host bits (n)
Address utilization	Wasteful every subnet reserves space for the largest requirement	Efficient no subnet reserves more space than it needs
Allocation order	Not order-dependent; subnets are simply equal-sized blocks	Allocated largest-to-smallest to avoid overlap
Complexity	Simple to plan and document	Requires more careful planning per subnet
Real-world use	Small, simple networks with similar-sized segments	Standard approach in enterprise and ISP networks

2.4 Subnetting Calculation Method (short bullet points)

Determine host bits needed: smallest n such that $2^n - 2 \geq \text{required hosts}$

New mask = 32 – n, or borrow bits so that $2^s \geq \text{required subnets}$

Block size = $2^{\text{host bits}}$

Network address = first address of block; Broadcast = last address of block

Usable range = network+1 to broadcast–1

2.5 Network Address and Broadcast Address

Network Address: The network address is the first address in a subnet's range. It identifies the subnet itself and cannot be assigned to any individual device. It is calculated by setting all host bits to 0.

Broadcast Address: The broadcast address is the last address in a subnet's range. It is used to send data to all devices within that subnet simultaneously and, like the network address, cannot be assigned to an individual device. It is calculated by setting all host bits to 1.

Because both the network address and the broadcast address are reserved and cannot be used by hosts, the number of usable host addresses in any subnet is always the block size minus 2.

Section 3: FLSM Worksheet 10 Solved Problems:

Method used for every problem:

- (1) find bits to borrow so $2^s \geq \text{required subnets}$
- (2) compute new prefix and block size
- (3) list subnets by adding the block size to the network address
- (4) usable range = network+1 to broadcast-1.

Problem 1: 192.168.5.0/24 → 4 equal subnets

Step 1 – Subnets needed: 4. Smallest s where $2^s \geq 4 \rightarrow s = 2$ bits borrowed.

Step 2 – New prefix = $/24 + 2 = /26$. Host bits remaining = 6.

Step 3 – Block size = $2^6 = 64$. Usable hosts = $64 - 2 = 62$.

Step 4 – Subnets (starting at 192.168.5.0):

FLSM Problem 1: 192.168.5.0/24 -> 4 subnets (/26, block 64, 62 usable hosts each)			
Subnet #	Network Address	Broadcast Address	Usable Host Range
1	192.168.5.0	192.168.5.63	192.168.5.1 - 192.168.5.62
2	192.168.5.64	192.168.5.127	192.168.5.65 - 192.168.5.126
3	192.168.5.128	192.168.5.191	192.168.5.129 - 192.168.5.190
4	192.168.5.192	192.168.5.255	192.168.5.193 - 192.168.5.254

Problem 2: 192.168.15.0/24 → 8 equal subnets

Step 1 – $2^s \geq 8 \rightarrow s = 3$ bits borrowed.

Step 2 – New prefix = $/24 + 3 = /27$. Host bits remaining = 5.

Step 3 – Block size = $2^5 = 32$. Usable hosts = 30.

Step 4 – Subnets (starting at 192.168.15.0):

FLSM Problem 2: 192.168.15.0/24 -> 8 subnets (/27, block 32, 30 usable hosts each)			
Subnet #	Network Address	Broadcast Address	Usable Host Range
1	192.168.15.0	192.168.15.31	192.168.15.1 - 192.168.15.30
2	192.168.15.32	192.168.15.63	192.168.15.33 - 192.168.15.62
3	192.168.15.64	192.168.15.95	192.168.15.65 - 192.168.15.94
4	192.168.15.96	192.168.15.127	192.168.15.97 - 192.168.15.126
5	192.168.15.128	192.168.15.159	192.168.15.129 - 192.168.15.158
6	192.168.15.160	192.168.15.191	192.168.15.161 - 192.168.15.190
7	192.168.15.192	192.168.15.223	192.168.15.193 - 192.168.15.222
8	192.168.15.224	192.168.15.255	192.168.15.225 - 192.168.15.254

Problem 3: 172.18.0.0/16 → 4 equal subnets

Step 1 – $2^s \geq 4 \rightarrow s = 2$ bits borrowed.

Step 2 – New prefix = /16 + 2 = /18. Host bits remaining = 14.

Step 3 – Block size = $2^{14} = 16384$, which is a jump of 64 in the 3rd octet ($16384 \div 256 = 64$). Usable hosts = 16382.

Step 4 – Subnets:

(FLSM Problem 3: 172.18.0.0/16 -> 4 subnets /18, block 16384, 16382 usable hosts each)			
Subnet #	Network Address	Broadcast Address	Usable Host Range
1	172.18.0.0/18	172.18.63.255	172.18.0.1 - 172.18.63.254
1	172.18.0.0/18	172.18.63.255	172.18.0.1 - 172.18.63.254
2	172.18.64.0/18	172.18.127.255	172.18.64.1 - 172.18.127.254
3	172.18.128.0/18	172.18.191.255	172.18.128.1 - 172.18.191.254
4	172.18.192.0/18	172.18.255.255	172.18.192.1 - 172.18.255.254

Problem 4: 10.20.0.0/16 → 16 equal subnets

Step 1 – $2^s \geq 16 \rightarrow s = 4$ bits borrowed.

Step 2 – New prefix = /16 + 4 = /20. Host bits remaining = 12.

Step 3 – Block size = $2^{12} = 4096$, a jump of 16 in the 3rd octet ($4096 \div 256 = 16$). Usable hosts = 4094.

Step 4 – Subnets (first 3 of 16 shown, pattern repeats):

Subnet #	Network Address	Broadcast Address	Usable Host Range
1	10.20.0.0/20	10.20.15.255	10.20.0.1 - 10.20.15.254
2	10.20.16.0/20	10.20.31.255	10.20.16.1 - 10.20.31.254
3	10.20.32.0/20	10.20.47.255	10.20.32.1 - 10.20.47.254
4	10.20.48.0/20	10.20.63.255	10.20.48.1 - 10.20.63.254
5	10.20.64.0/20	10.20.79.255	10.20.64.1 - 10.20.79.254
6	10.20.80.0/20	10.20.95.255	10.20.80.1 - 10.20.95.254
7	10.20.96.0/20	10.20.111.255	10.20.96.1 - 10.20.111.254
8	10.20.112.0/20	10.20.127.255	10.20.112.1 - 10.20.127.254
9	10.20.128.0/20	10.20.143.255	10.20.128.1 - 10.20.143.254
10	10.20.144.0/20	10.20.159.255	10.20.144.1 - 10.20.159.254
11	10.20.160.0/20	10.20.175.255	10.20.160.1 - 10.20.175.254
12	10.20.176.0/20	10.20.191.255	10.20.176.1 - 10.20.191.254
13	10.20.192.0/20	10.20.207.255	10.20.192.1 - 10.20.207.254
14	10.20.208.0/20	10.20.223.255	10.20.208.1 - 10.20.223.254
15	10.20.224.0/20	10.20.239.255	10.20.224.1 - 10.20.239.254
16	10.20.240.0/20	10.20.255.255	10.20.240.1 - 10.20.255.254

Problem 5: 192.168.25.0/24 → 2 equal subnets

Step 1 – $2^s \geq 2 \rightarrow s = 1$ bit borrowed.

Step 2 – New prefix = /24 + 1 = /25. Host bits remaining = 7.

Step 3 – Block size = $2^7 = 128$. Usable hosts = 126.

Step 4 – Subnets:

Subnet #	Network Address	Broadcast Address	Usable Host Range
1	192.168.25.0	192.168.25.127	192.168.25.1 - 192.168.25.126
2	192.168.25.128	192.168.25.255	192.168.25.129 - 192.168.25.254

Problem 6: 192.168.60.0/24 → 32 equal subnets

Step 1 – $2^s \geq 32 \rightarrow s = 5$ bits borrowed.

Step 2 – New prefix = $/24 + 5 = /29$. Host bits remaining = 3.

Step 3 – Block size = $2^3 = 8$. Usable hosts = 6.

Step 4 – Subnets (first 3 of 32 shown):

Subnet #	Network Address	Broadcast Address	Usable Host Range
1	192.168.60.0	192.168.60.7	192.168.60.1 - 192.168.60.6
2	192.168.60.8	192.168.60.15	192.168.60.9 - 192.168.60.14
3	192.168.60.16	192.168.60.23	192.168.60.17 - 192.168.60.22
4	192.168.60.24	192.168.60.31	192.168.60.25 - 192.168.60.30
5	192.168.60.32	192.168.60.39	192.168.60.33 - 192.168.60.38
6	192.168.60.40	192.168.60.47	192.168.60.41 - 192.168.60.46
7	192.168.60.48	192.168.60.55	192.168.60.49 - 192.168.60.54
8	192.168.60.56	192.168.60.63	192.168.60.57 - 192.168.60.62
9	192.168.60.64	192.168.60.71	192.168.60.65 - 192.168.60.70
10	192.168.60.72	192.168.60.79	192.168.60.73 - 192.168.60.78
11	192.168.60.80	192.168.60.87	192.168.60.81 - 192.168.60.86
12	192.168.60.88	192.168.60.95	192.168.60.89 - 192.168.60.94
13	192.168.60.96	192.168.60.103	192.168.60.97 - 192.168.60.102
14	192.168.60.104	192.168.60.111	192.168.60.105 - 192.168.60.110
15	192.168.60.112	192.168.60.119	192.168.60.113 - 192.168.60.118
16	192.168.60.120	192.168.60.127	192.168.60.121 - 192.168.60.126
17	192.168.60.128	192.168.60.135	192.168.60.129 - 192.168.60.134
18	192.168.60.136	192.168.60.143	192.168.60.137 - 192.168.60.142
19	192.168.60.144	192.168.60.151	192.168.60.145 - 192.168.60.150
20	192.168.60.152	192.168.60.159	192.168.60.153 - 192.168.60.158
21	192.168.60.160	192.168.60.167	192.168.60.161 - 192.168.60.166
22	192.168.60.168	192.168.60.175	192.168.60.169 - 192.168.60.174
23	192.168.60.176	192.168.60.183	192.168.60.177 - 192.168.60.182
24	192.168.60.184	192.168.60.191	192.168.60.185 - 192.168.60.190
25	192.168.60.192	192.168.60.199	192.168.60.193 - 192.168.60.198
26	192.168.60.200	192.168.60.207	192.168.60.201 - 192.168.60.206
27	192.168.60.208	192.168.60.215	192.168.60.209 - 192.168.60.214
28	192.168.60.216	192.168.60.223	192.168.60.217 - 192.168.60.222
29	192.168.60.224	192.168.60.231	192.168.60.225 - 192.168.60.230
30	192.168.60.232	192.168.60.239	192.168.60.233 - 192.168.60.238
31	192.168.60.240	192.168.60.247	192.168.60.241 - 192.168.60.246
32	192.168.60.248	192.168.60.255	192.168.60.249 - 192.168.60.254

Problem 7: 172.25.0.0/16 → 8 equal subnets

Step 1 – $2^s \geq 8 \rightarrow s = 3$ bits borrowed.

Step 2 – New prefix = $/16 + 3 = /19$. Host bits remaining = 13.

Step 3 – Block size = $2^{13} = 8192$, a jump of 32 in the 3rd octet. Usable hosts = 8190.

Step 4 – Subnets (first 3 of 8 shown):

Subnet #	Network Address	Broadcast Address	Usable Host Range
1	172.25.0.0/19	172.25.31.255	172.25.0.1 - 172.25.31.254
2	172.25.32.0/19	172.25.63.255	172.25.32.1 - 172.25.63.254
3	172.25.64.0/19	172.25.95.255	172.25.64.1 - 172.25.95.254
4	172.25.96.0/19	172.25.127.255	172.25.96.1 - 172.25.127.254
5	172.25.128.0/19	172.25.159.255	172.25.128.1 - 172.25.159.254
6	172.25.160.0/19	172.25.191.255	172.25.160.1 - 172.25.191.254
7	172.25.192.0/19	172.25.223.255	172.25.192.1 - 172.25.223.254
8	172.25.224.0/19	172.25.255.255	172.25.224.1 - 172.25.255.254

Problem 8: 10.30.0.0/16 → 4 equal subnets

Step 1 – Subnets needed: 4. Smallest s where $2^s \geq 4 \rightarrow s = 2$ bits borrowed.

Step 2 – New prefix = $/16 + 2 = /18$. Host bits remaining = 14.

Step 3 – Block size = $2^{14} = 16384$, a jump of 64 in the 3rd octet ($16384 \div 256 = 64$). Usable hosts = 16382.

Step 4 – Subnets:

Subnet #	Network Address	Broadcast Address	Usable Host Range
1	10.30.0.0/18	10.30.63.255	10.30.0.1 - 10.30.63.254
2	10.30.64.0/18	10.30.127.255	10.30.64.1 - 10.30.127.254
3	10.30.128.0/18	10.30.191.255	10.30.128.1 - 10.30.191.254
4	10.30.192.0/18	10.30.255.255	10.30.192.1 - 10.30.255.254

Problem 9: 192.168.110.0/24 → 6 equal subnets

Step 1 – $2^s \geq 6 \rightarrow s = 3$ bits borrowed (8 subnets become available, only 6 needed).

Step 2 – New prefix = $/24 + 3 = /27$. Host bits remaining = 5.

Step 3 – Block size = $2^5 = 32$. Usable hosts = 30.

Step 4 – First 6 of the 8 available subnets are used:

Subnet #	Network Address	Broadcast Address	Usable Host Range
1	192.168.110.0	192.168.110.31	192.168.110.1 - 192.168.110.30
2	192.168.110.32	192.168.110.63	192.168.110.33 - 192.168.110.62
3	192.168.110.64	192.168.110.95	192.168.110.65 - 192.168.110.94
4	192.168.110.96	192.168.110.127	192.168.110.97 - 192.168.110.126
5	192.168.110.128	192.168.110.159	192.168.110.129 - 192.168.110.158
6	192.168.110.160	192.168.110.191	192.168.110.161 - 192.168.110.190

Problem 10: 172.28.0.0/16 → 6 equal subnets

Step 1 – Subnets needed: 6. Smallest s where $2^s \geq 6 \rightarrow s = 3$ bits borrowed ($2^3 = 8$ subnets become available, 6 needed).

Step 2 – New prefix = $/16 + 3 = /19$. Host bits remaining = 13.

Step 3 – Block size = $2^{13} = 8192$, a jump of 32 in the 3rd octet. Usable hosts = 8190.

Step 4 – First 6 of the 8 available subnets are used:

Subnet #	Network Address	Broadcast Address	Usable Host Range
1	172.28.0.0/19	172.28.31.255	172.28.0.1 - 172.28.31.254
2	172.28.32.0/19	172.28.63.255	172.28.32.1 - 172.28.63.254
3	172.28.64.0/19	172.28.95.255	172.28.64.1 - 172.28.95.254
4	172.28.96.0/19	172.28.127.255	172.28.96.1 - 172.28.127.254

5	172.28.128.0/19	172.28.159.255	172.28.128.1 - 172.28.159.254
6	172.28.160.0/19	172.28.191.255	172.28.160.1 - 172.28.191.254

Section 4: VLSM Worksheet 5 Design Scenarios

Method for every scenario:

1. Sort all host requirements from **largest to smallest**
2. For each requirement, find the smallest **n** where $2^n - 2 \geq$ hosts needed
3. Assign the next available block of size 2^n , starting right where the previous block ended
4. Calculate mask, network address, broadcast address, and usable range

Scenario 1 Base network: 192.168.2.0/24

Requirements: Admin 60, Finance 28, Networking 12, IoT Devices 6, Router Link 2

Sorted largest→smallest: 60, 28, 12, 6, 2

Admin (60 hosts):

- Find n: $2^n - 2 \geq 60 \rightarrow n = 6$ ($2^6 - 2 = 62$)
- Mask = /26, block size = 64
- Assign: **192.168.2.0/26** → Broadcast: .63 → Usable: .1 – .62

Finance (28 hosts):

- $n = 5$ ($2^5 - 2 = 30$)
- Mask = /27, block size = 32
- Starts right after Admin's block ended (.64)
- Assign: **192.168.2.64/27** → Broadcast: .95 → Usable: .65 – .94

Networking (12 hosts):

- $n = 4$ ($2^4 - 2 = 14$)
- Mask = /28, block size = 16
- Starts after Finance (.96)
- Assign: **192.168.2.96/28** → Broadcast: .111 → Usable: .97 – .110

IoT Devices (6 hosts):

- $n = 3$ ($2^3 - 2 = 6$)
- Mask = /29, block size = 8
- Starts after Networking (.112)
- Assign: **192.168.2.112/29** → Broadcast: .119 → Usable: .113 – .118

Router Link (2 hosts):

- $n = 2$ ($2^2 - 2 = 2$)
- Mask = /30, block size = 4
- Starts after IoT (.120)
- Assign: **192.168.2.120/30** → Broadcast: .123 → Usable: .121 – .122

Req	n(host bits)	Mask	Assigned Subnet	Network	Broadcast	Usable Range
Admin	6	/26	192.168.2.0 /26	.0	.63	.1 - .62
Finance	5	/27	192.168.2.64 /27	.64	.95	.65 - .94
Networking	4	/28	192.168.2.96 /28	.96	.111	.97 - .110
IoT Devices	3	/29	192.168.2.112 /29	.112	.119	.113 - .118
Router Link	2	/30	192.168.2.120 /30	.120	.123	.121 - .122

Scenario 2 Base network: 10.1.0.0/24

Requirements: Production 90, QA 40, Support 20, Reception 8, Link 2

Sorted: 90, 40, 20, 8, 2

Production (90 hosts):

- $n = 7$ ($2^7 - 2 = 126$)
- Mask = /25, block size = 128
- Assign: **10.1.0.0/25** → Broadcast: .127 → Usable: .1 – .126

QA (40 hosts):

- $n = 6$ ($2^6 - 2 = 62$)
- Mask = /26, block size = 64. Starts at .128
- Assign: **10.1.0.128/26** → Broadcast: .191 → Usable: .129 – .190

Support (20 hosts):

- $n = 5$ ($2^5 - 2 = 30$)
- Mask = /27, block size = 32. Starts at .192
- Assign: **10.1.0.192/27** → Broadcast: .223 → Usable: .193 – .222

Reception (8 hosts):

- $n = 4$ ($2^4 - 2 = 14$)
- Mask = /28, block size = 16. Starts at .224
- Assign: **10.1.0.224/28** → Broadcast: .239 → Usable: .225 – .238

Link (2 hosts):

- $n = 2$ ($2^2 - 2 = 2$)
- Mask = /30, block size = 4. Starts at .240
- Assign: **10.1.0.240/30** → Broadcast: .243 → Usable: .241 – .242

Requirement	Mask	Assigned Subnet	Network	Broadcast	Usable Range
Production	/25	10.1.0.0/25	.0	.127	.1 - .126
QA	/26	10.1.0.128/26	.128	.191	.129 - .190
Support	/27	10.1.0.192/27	.192	.223	.193 - .222
Reception	/28	10.1.0.224/28	.224	.239	.225 - .238
Link	/30	10.1.0.240/30	.240	.243	.241 - .242

Scenario 3 Base network: 172.17.0.0/22

This base is bigger /22 gives 1024 total addresses (172.17.0.0 – 172.17.3.255), needed because HQ alone requires 400 hosts.

Requirements: HQ 400, Branch-A 180, Branch-B 90, Branch-C 40, WAN Link 2

Sorted: 400, 180, 90, 40, 2

HQ (400 hosts):

- $n = 9$ ($2^9 - 2 = 510$) ($n=8$ would only give 254, not enough)
- Mask = /23, block size = 512
- Assign: **172.17.0.0/23** → Broadcast: 172.17.1.255 → Usable: 172.17.0.1 – 172.17.1.254

Branch-A (180 hosts):

- $n = 8$ ($2^8 - 2 = 254$)
- Mask = /24, block size = 256. Starts at 172.17.2.0
- Assign: **172.17.2.0/24** → Broadcast: 172.17.2.255 → Usable: 172.17.2.1 – 172.17.2.254

Branch-B (90 hosts):

- $n = 7$ ($2^7 - 2 = 126$)
- Mask = /25, block size = 128. Starts at 172.17.3.0
- Assign: **172.17.3.0/25** → Broadcast: 172.17.3.127 → Usable: 172.17.3.1 – 172.17.3.126

Branch-C (40 hosts):

- $n = 6$ ($2^6 - 2 = 62$)
- Mask = /26, block size = 64. Starts at 172.17.3.128
- Assign: **172.17.3.128/26** → Broadcast: 172.17.3.191 → Usable: 172.17.3.129 – 172.17.3.190

WAN Link (2 hosts):

- $n = 2$. Mask = /30, block size = 4. Starts at 172.17.3.192
- Assign: **172.17.3.192/30** → Broadcast: 172.17.3.195 → Usable: 172.17.3.193 – 172.17.3.194

Req	Mask	Assigned Subnet	Network	Broadcast	Usable Range
HQ	/23	172.17.0.0/23	172.17.0.0	172.17.1.255	172.17.0.1 - 172.17.1.254
Branch-A	/24	172.17.2.0/24	172.17.2.0	172.17.2.255	172.17.2.1 - 172.17.2.254
Branch-B	/25	172.17.3.0/25	172.17.3.0	172.17.3.127	172.17.3.1 - 172.17.3.126
Branch-C	/26	172.17.3.128/26	172.17.3.128	172.17.3.191	172.17.3.129 - 172.17.3.190
WAN Link	/30	172.17.3.192/30	172.17.3.192	172.17.3.195	172.17.3.193 - 172.17.3.194

Scenario 4 Base network: 192.168.70.0/24

Requirements: Design 35, Testing 18, Server Room 8, Kiosk 4, Link 2

Sorted: 35, 18, 8, 4, 2

Design (35 hosts):

- $n = 6$ ($2^6 - 2 = 62$)
- Mask = /26, block size = 64
- Assign: **192.168.70.0/26** → Broadcast: .63 → Usable: .1 – .62

Testing (18 hosts):

- $n = 5$ ($2^5 - 2 = 30$)
- Mask = /27, block size = 32. Starts at .64
- Assign: **192.168.70.64/27** → Broadcast: .95 → Usable: .65 – .94

Server Room (8 hosts):

- $n = 4$ ($2^4 - 2 = 14$)
- Mask = /28, block size = 16. Starts at .96
- Assign: **192.168.70.96/28** → Broadcast: .111 → Usable: .97 – .110

Kiosk (4 hosts):

- $n = 3$ ($2^3 - 2 = 6$)
- Mask = /29, block size = 8. Starts at .112
- Assign: **192.168.70.112/29** → Broadcast: .119 → Usable: .113 – .118

Link (2 hosts):

- $n = 2$. Mask = /30, block size = 4. Starts at .120
- Assign: **192.168.70.120/30** → Broadcast: .123 → Usable: .121 – .122

Req	Mask	Assigned Subnet	Network	Broadcast	Usable Range
Design	/26	192.168.70.0/26	.0	.63	.1 - .62
Testing	/27	192.168.70.64/27	.64	.95	.65 - .94
Server Room	/28	192.168.70.96/28	.96	.111	.97 - .110
Kiosk	/29	192.168.70.112/29	.112	.119	.113 - .118
Link	/30	192.168.70.120/30	.120	.123	.121 - .122

Scenario 5 Base network: 192.168.210.0/24

Requirements: Marketing 40, Warehouse 20, Security 8

Sorted: 40, 20, 8

Marketing (40 hosts):

$n=6$ ($2^6-2=62$)

Mask /26, block 64

Assign: 192.168.210.0/26

Broadcast: .63 | Usable: .1-.62

Warehouse (20 hosts):

$n=5$ ($2^5-2=30$)

Mask /27, block 32, starts at .64

Assign: 192.168.210.64/27

Broadcast: .95 | Usable: .65-.94

Security (8 hosts):

$n=4$ ($2^4-2=14$)

Mask /28, block 16, starts at .96

Assign: 192.168.210.96/28

Broadcast: .111 | Usable: .97-.110

Requirement	Mask	Assigned Subnet	Network	Broadcast	Usable Range
Marketing	/26	192.168.210.0/26	.0	.63	.1 - .62
Warehouse	/27	192.168.210.64/27	.64	.95	.65 - .94
Security	/28	192.168.210.96/28	.96	.111	.97 - .110

Section 5: VLAN Configuration in Cisco Packet Tracer

5.1 Topology

SW1 Fa0/1 → PC1 (VLAN 10 – Marketing, 192.168.210.10 /26)

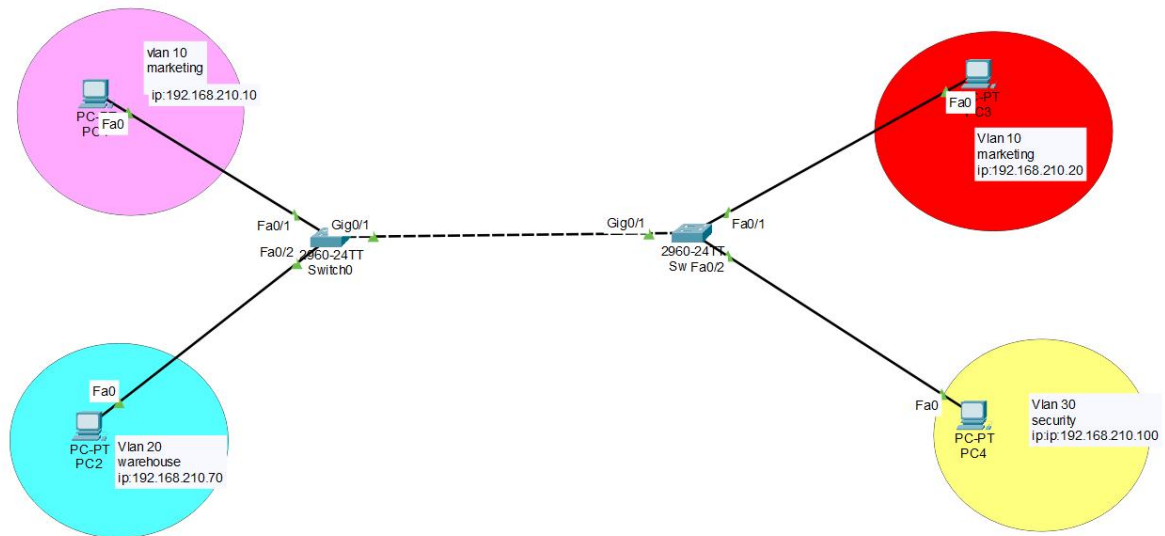
SW1 Fa0/2 → PC2 (VLAN 20 – Warehouse, 192.168.210.70 /27)

SW1 Gi0/1 → trunk to SW2

SW2 Fa0/1 → PC3 (VLAN 10 – Marketing, 192.168.210.20 /26)

SW2 Fa0/2 → PC4 (VLAN 30 – Security, 192.168.210.100 /28)

SW2 Gi0/1 → trunk to SW1



5.2 SW1 Configuration

```
enable
configure terminal
hostname SW1
vlan 10
name Marketing
vlan 20
name Warehouse
vlan 30
name Security
exit
interface fastEthernet 0/1
switchport mode access
switchport access vlan 10
exit
interface fastEthernet 0/2
switchport mode access
switchport access vlan 20
exit
interface gigabitEthernet 0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30
exit
end
Copy running-config startup-config
```

5.3 SW2 Configuration

```

Switch>enable
Switch#config ter
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW2
SW2(config)#vlan 10
SW2(config-vlan)#name marketing
SW2(config-vlan)#vlan 20
SW2(config-vlan)#name warehouse
SW2(config-vlan)#vlan 30
SW2(config-vlan)#name security
SW2(config-vlan)#exit
SW2(config)#interface fastethernet 0/1
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 10
SW2(config-if)#exit
SW2(config)#interface fastethernet 0/2
SW2(config-if)#switchport mode access
SW2(config-if)#switchport access vlan 30
SW2(config-if)#exit
SW2(config)#interface gigabitethernet 0/1
SW2(config-if)#switchport mode trunk
SW2(config-if)#switchport trunk allowed vlan 10,20,30
SW2(config-if)#exit
SW2(config)#exit
SW2#
%SYS-5-CONFIG_I: Configured from console by console

SW2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]

```

5.4 Verification SW1

```
SW1>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10	marketing	active	Fa0/1
20	warehouse	active	Fa0/2
30	security	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW1>show interfaces trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gig0/1	10,20,30

Port	Vlans allowed and active in management domain
Gig0/1	10,20,30

Port	Vlans in spanning tree forwarding state and not pruned
Gig0/1	10,20,30

```
SW1>show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	down	down
FastEthernet0/4	unassigned	YES	manual	down	down
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down
FastEthernet0/22	unassigned	YES	manual	down	down
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	manual	down	down

5.5 Verification SW2

```
SW2>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/3, Fa0/4, Fa0/5, Fa0/6 Fa0/7, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/2
10	marketing	active	Fa0/1
20	warehouse	active	
30	security	active	Fa0/2
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW2>show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Gig0/1	10,20,30			
Port	Vlans allowed and active in management domain			
Gig0/1	10,20,30			
Port	Vlans in spanning tree forwarding state and not pruned			
Gig0/1	10,20,30			

```
SW2>show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	down	down
FastEthernet0/4	unassigned	YES	manual	down	down
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down
FastEthernet0/22	unassigned	YES	manual	down	down
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	manual	down	down
Vlan1	unassigned	YES	manual	administratively down	down

5.6 PC Configuration

PC	VLAN	IP Address	Subnet Mask
PC1	10 Marketing	192.168.210.10	255.255.255.192
PC2	20 Warehouse	192.168.210.70	255.255.255.224
PC3	10 Marketing	192.168.210.20	255.255.255.192
PC4	30 Security	192.168.210.100	255.255.255.240

5.7 ping Results

PC1

PC3

```
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.210.20

Pinging 192.168.210.20 with 32 bytes of data:

Reply from 192.168.210.20: bytes=32 time=7ms TTL=128
Reply from 192.168.210.20: bytes=32 time<1ms TTL=128
Reply from 192.168.210.20: bytes=32 time=5ms TTL=128
Reply from 192.168.210.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.210.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 3ms

C:\>ping 192.168.210.100

Pinging 192.168.210.100 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.210.100:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

PC4

```
PC3
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.210.10

Pinging 192.168.210.10 with 32 bytes of data:

Reply from 192.168.210.10: bytes=32 time<1ms TTL=128
Reply from 192.168.210.10: bytes=32 time<1ms TTL=128
Reply from 192.168.210.10: bytes=32 time<1ms TTL=128
Reply from 192.168.210.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.210.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.210.70

Pinging 192.168.210.70 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.210.70:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

```
PC4
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 19.168.210.20

Pinging 19.168.210.20 with 32 bytes of data:

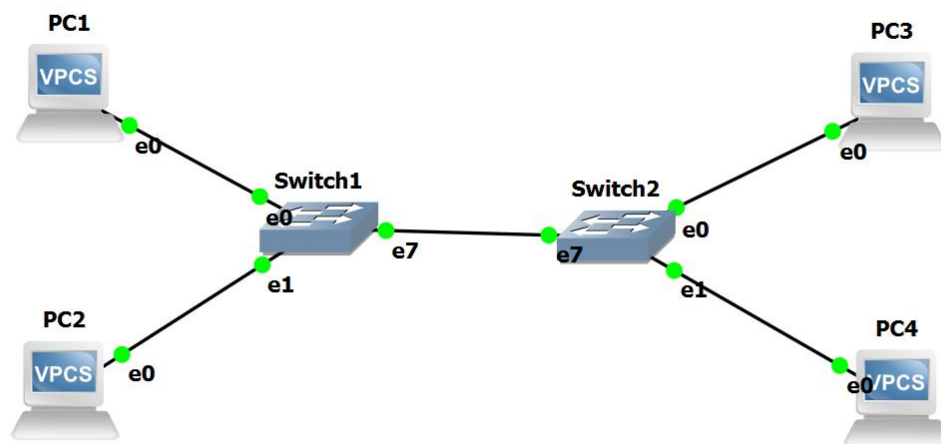
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 19.168.210.20:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Section 6: VLAN Configuration in GNS3

6.1 Topology

Same 2 switches + 4 PCs, connected the same way as Packet Tracer above.



6.2 Port-to-VLAN Mapping

SW1: Port 0 = VLAN 10 (Access) | Port 1 = VLAN 20 (Access) | Port 7 = VLAN 1 (Dot1q trunk)

SW2: Port 0 = VLAN 10 (Access) | Port 1 = VLAN 30 (Access) | Port 7 = VLAN 1 (Dot1q trunk)

Switch1 configuration

General

Name: Switch1

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Add

Delete

Ports

Port	VLAN	Type	EtherType
0	10	access	
1	20	access	
2	1	access	
3	1	access	
4	1	access	
5	1	access	
6	1	access	
7	1	dot1q	

Switch2 configuration

General

Name: Switch2

Console type: none

Settings

Port: 8

VLAN: 1

Type: access

QinQ EtherType: 0x8100

Add

Delete

Ports

Port	VLAN	Type	EtherType
0	10	access	
1	30	access	
2	1	access	
3	1	access	
4	1	access	
5	1	access	
6	1	access	
7	1	dot1q	

6.3 PC (VPCS) IPs

PC1> ip 192.168.210.10/26

PC2> ip 192.168.210.70/27

PC3> ip 192.168.210.20/26

PC4> ip 192.168.210.100/28

6.4 Ping Test

PC1

```
PC1 PC2 PC3
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 192.168.210.10/26
Checking for duplicate address...
PC1 : 192.168.210.10 255.255.255.192

PC1> ping 192.168.210.20
64 bytes from 192.168.210.20 icmp_seq=1 ttl=64 time=0.915 ms
64 bytes from 192.168.210.20 icmp_seq=2 ttl=64 time=1.332 ms
64 bytes from 192.168.210.20 icmp_seq=3 ttl=64 time=1.334 ms
64 bytes from 192.168.210.20 icmp_seq=4 ttl=64 time=1.194 ms
64 bytes from 192.168.210.20 icmp_seq=5 ttl=64 time=0.983 ms

PC1> 
```

PC2:

```

Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC2> ip 192.168.210.70/27
Checking for duplicate address...
PC1 : 192.168.210.70 255.255.255.224

PC2> ping 192.168.210.1
No gateway found

PC2> ping 192.168.210.10
No gateway found

PC2> █
```

PC3

```

Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC3> ip 192.168.210.20/26
Checking for duplicate address...
PC1 : 192.168.210.20 255.255.255.192

PC3> ping 192.168.210.10
64 bytes from 192.168.210.10 icmp_seq=1 ttl=64 time=0.759 ms
64 bytes from 192.168.210.10 icmp_seq=2 ttl=64 time=1.073 ms
64 bytes from 192.168.210.10 icmp_seq=3 ttl=64 time=1.442 ms
64 bytes from 192.168.210.10 icmp_seq=4 ttl=64 time=1.022 ms
64 bytes from 192.168.210.10 icmp_seq=5 ttl=64 time=1.044 ms

PC3> █
```

PC4



PC1

PC2

PC3

PC4



```
Welcome to Virtual PC Simulator, version 0.6.2
Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.
```

```
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Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.
```

```
Press '?' to get help.
```

```
Executing the startup file
```

```
PC4> ip 192.168.210.100/28
Checking for duplicate address...
PC1 : 192.168.210.100 255.255.255.240
```

```
PC4> 
```